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Rstudio Packages

```
>library(lme4)
```

Loading required package: Matrix

Registered S3 method overwritten by 'lme4':

method from

na.action.merMod car

Warning message:

package ‘lme4’ was built under R version 4.4.3

```
> library(tidyverse)
```

— **Attaching core tidyverse packages**

— tidyverse 2.0.0 —

✓ dplyr 1.1.4 ✓ readr 2.1.5

✓ forcats 1.0.0 ✓ stringr 1.5.1

✓ ggplot2 3.5.1 ✓ tibble 3.2.1

✓ lubridate 1.9.4 ✓ tidyr 1.3.1

✓ purrr 1.0.2

— **Conflicts**

——————————— tidyverse_conflicts() —

✕ tidyr::expand() masks Matrix::expand()

✕ dplyr::filter() masks stats::filter()

✕ dplyr::lag() masks stats::lag()

✕ tidyr::pack() masks Matrix::pack()

✕ tidyr::unpack() masks Matrix::unpack()

ℹ Use the conflicted package to force all conflicts to become errors

Warning messages:

1: package ‘tidyverse’ was built under R version 4.4.3

2: package ‘readr’ was built under R version 4.4.3

3: package ‘dplyr’ was built under R version 4.4.3

4: package ‘forcats’ was built under R version 4.4.3

5: package ‘lubridate’ was built under R version 4.4.3

```
> library(emmeans)
```

Welcome to emmeans.

Caution: You lose important information if you filter this package's results.

See '? untidy'

Warning message:

package ‘emmeans’ was built under R version 4.4.3

```
> library(performance)
```

Warning message: package ‘performance’ was built under R version 4.4.3

```
> library(DHARMa)
```

This is DHARMa 0.5.0. For overview type '?DHARMa'. For recent changes, type news(package = 'DHARMa')

Note that the default setting in simulateResiduals was changed to conditional simulations since version 0.5.0. This is likely to change residual calculations for all hierarchical (in particular random effect) models. If you want to switch back to the old package version defaults, please use the argument simulateREs = "user-specified" in simulateResiduals(). For more details, see ?simulateREsiduals.

Attaching package: ‘DHARMa’

The following object is masked from ‘package:lme4’:

getData

Read Data

```
> checkN <- function(data,label){
+ print(paste(label,": ",NROW(data),sep = ""))
+ return(data)
+ }
> fc<-read.csv("C:/Users/sarah/Downloads/Nieuwe map/fulldata_changes.csv", sep = ";")
> checkN(fc, "Data_wide_format")
[1] " Data_wide_format: 855"
```

ResponseId Progress Gender Sex Age SO dating_exp Status.1 BFI1_1 BFI1_2 BFI1_3 BFI1_4 BFI1_5 BFI1_6 B

1 R_BLKJUB3sKlfLvaN 100 1 1 21 1 1 2 5 2 5 4 4 3 3 5 2 5

2 R_2b45x8P15jBbZSK 100 1 1 20 2 1 1 4 2 4 2 4 4 4 4 3 5

BFI1_11 BFI1_12 BFI1_13 BFI1_14 BFI1_15 BFI1_16 BFI1_17 BFI1_18 BFI1_19 BFI1_20 BFI1_21 BFI1_22 BFI1_2

1 4 1 3 2 4 4 5 5 2 5 2 5 4 1 5 4 4

2 2 2 4 2 3 2 5 4 5 4 2 4 4 4 2 4 2

BFI1_28 BFI1_29 BFI1_30 BFI1_31 BFI1_32 BFI1_33 BFI1_34 BFI1_35 BFI1_36 BFI1_37 BFI1_38 BFI1_39 BFI1_4

1 2 4 2 4 5 4 4 5 5 4 5 2 4 4 4 5 5

2 4 4 4 4 4 4 3 4 4 2 4 5 4 4 3 5 4

Q10 X1_Q9 X2_Q9 X3_Q9 X4_Q9 X5_Q9 X6_Q9 X7_Q9 X8_Q9 X9_Q9 X10_Q9 X11_Q9 X12_Q9 X13_Q9 X14_Q9

1 2 1 2 2 2 2 1 1 1 1 1 1 1 2 1 2 1 1 1 2 1

2 1 NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA

X21_Q9 X22_Q9 X23_Q9 X24_Q9 X25_Q9 X26_Q9 X27_Q9 X28_Q9 X29_Q9 X30_Q9 X31_Q9 X32_Q9 X33_Q9 X3

1 1 1 1 1 2 2 1 2 2 1 2 1 1 2 1 1 2 1 2

2 NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA

X40_Q9 X41_Q9 X42_Q9 X43_Q9 X44_Q9 X45_Q9 X46_Q9 X47_Q9 X48_Q9 X49_Q9 X50_Q9 X51_Q9 X52_Q9 X5

1 1 1 1 2 1 1 1 2 1 1 1 2 1 2 1 1 1 2 1

2 NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA

X59_Q9 X60_Q9 X61_Q9 X62_Q9 X63_Q9 X64_Q9 X65_Q9 X66_Q9 X67_Q9 X68_Q9 X69_Q9 X70_Q9 X71_Q9 X7

1 1 1 2 1 1 1 2 1 1 1 1 1 1 1 1 1 1 2 2

2 NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA

X78_Q9 X79_Q9 X80_Q9 X81_Q9 X82_Q9 X83_Q9 X84_Q9 X85_Q9 X86_Q9 X87_Q9 X88_Q9 X89_Q9 X90_Q9 X9

1 1 1 1 1 2 1 1 1 2 1 1 1 1 1 2 1 1 1 2

2 NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA

X97_Q9 X98_Q9 X99_Q9 X100_Q9 X1_Q13 X2_Q13 X3_Q13 X4_Q13 X5_Q13 X6_Q13 X7_Q13 X8_Q13 X9_Q13 X

1 1 2 1 1 NA NA NA NA NA NA NA NA NA NA NA NA NA NA

2 NA NA NA NA 2 1 2 2 2 1 2 2 1 1 2 2 1 2

X15_Q13 X16_Q13 X17_Q13 X18_Q13 X19_Q13 X20_Q13 X21_Q13 X22_Q13 X23_Q13 X24_Q13 X25_Q13 X26_

1 NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA

2 2 2 2 2 1 2 2 1 1 2 1 1 2 2 1 2 2

X32_Q13 X33_Q13 X34_Q13 X35_Q13 X36_Q13 X37_Q13 X38_Q13 X39_Q13 X40_Q13 X41_Q13 X42_Q13 X43_

1 NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA

2 2 2 2 2 2 2 2 2 2 2 2 2 2 1 1 2 2

X49_Q13 X50_Q13 X51_Q13 X52_Q13 X53_Q13 X54_Q13 X55_Q13 X56_Q13 X57_Q13 X58_Q13 X59_Q13 X60_

1 NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA

2 2 2 2 1 2 2 2 2 2 2 2 1 2 1 1 2 1

X66_Q13 X67_Q13 X68_Q13 X69_Q13 X70_Q13 X71_Q13 X72_Q13 X73_Q13 X74_Q13 X75_Q13 X76_Q13 X77_

1 NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA

2 1 2 1 2 2 2 2 1 2 2 2 2 2 1 2 2 1

X83_Q13 X84_Q13 X85_Q13 X86_Q13 X87_Q13 X88_Q13 X89_Q13 X90_Q13 X91_Q13 X92_Q13 X93_Q13 X94_

1 NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA

2 2 1 2 1 2 2 2 2 1 1 1 2 1 2 2 2 1

X100_Q13 Desc_or_pic Attention Q20 Q21 Random.ID image1 image2 image3 image4 image5 image6 ima

1 NA 3 1 2 6476 IMAGE1 IMAGE51 IMAGE132 IMAGE39 IMAGE111 IMAGE10 IMAGE3 IMAGE52

2 2 1 1 2 6934 IMAGE2 IMAGE13 IMAGE81 IMAGE19 IMAGE26 IMAGE119 IMAGE195 IMAGE67

image11 image12 image13 image14 image15 image16 image17 image18 image19 image20 image21 imag

1 IMAGE54 IMAGE106 IMAGE168 IMAGE36 IMAGE112 IMAGE108 IMAGE134 IMAGE200 IMAGE194 IMAGE164 I

2 IMAGE92 IMAGE69 IMAGE30 IMAGE57 IMAGE78 IMAGE63 IMAGE135 IMAGE82 IMAGE158 IMAGE113 IMAG

image26 image27 image28 image29 image30 image31 image32 image33 image34 image35 image36 image

1 IMAGE80 IMAGE180 IMAGE33 IMAGE162 IMAGE22 IMAGE146 IMAGE144 IMAGE62 IMAGE141 IMAGE77 IMA

2 IMAGE138 IMAGE23 IMAGE89 IMAGE196 IMAGE95 IMAGE161 IMAGE42 IMAGE192 IMAGE167 IMAGE20 IMAG

image42 image43 image44 image45 image46 image47 image48 image49 image50 image51 image52 imag

1 IMAGE58 IMAGE148 IMAGE189 IMAGE187 IMAGE188 IMAGE120 IMAGE125 IMAGE126 IMAGE160 IMAGE173 I

2 IMAGE84 IMAGE5 IMAGE40 IMAGE72 IMAGE76 IMAGE35 IMAGE185 IMAGE182 IMAGE178 IMAGE137 IMAG

image57 image58 image59 image60 image61 image62 image63 image64 image65 image66 image67 imag

1 IMAGE147 IMAGE110 IMAGE115 IMAGE83 IMAGE136 IMAGE68 IMAGE46 IMAGE172 IMAGE60 IMAGE101 IMA

2 IMAGE14 IMAGE18 IMAGE174 IMAGE128 IMAGE142 IMAGE65 IMAGE109 IMAGE153 IMAGE159 IMAGE21 IMA

image72 image73 image74 image75 image76 image77 image78 image79 image80 image81 image82 imag

1 IMAGE71 IMAGE140 IMAGE166 IMAGE99 IMAGE181 IMAGE150 IMAGE48 IMAGE155 IMAGE86 IMAGE165 IM

2 IMAGE100 IMAGE127 IMAGE16 IMAGE169 IMAGE7 IMAGE79 IMAGE184 IMAGE104 IMAGE186 IMAGE121 IM

image87 image88 image89 image90 image91 image92 image93 image94 image95 image96 image97 image

1 IMAGE97 IMAGE114 IMAGE49 IMAGE156 IMAGE64 IMAGE145 IMAGE9 IMAGE91 IMAGE59 IMAGE123 IMAGE

2 IMAGE24 IMAGE129 IMAGE87 IMAGE25 IMAGE37 IMAGE41 IMAGE53 IMAGE50 IMAGE179 IMAGE15 IMAGE1

text3 text4 text5 text6 text7 text8 text9 text10 text11 text12 text13 text14 text15 text16 text17 text18 text19 t

1 TEXT71 TEXT84 TEXT78 TEXT95 TEXT4 TEXT34 TEXT69 TEXT13 TEXT46 TEXT91 TEXT55 TEXT2 TEXT60 TEXT83 T

2 TEXT91 TEXT90 TEXT32 TEXT75 TEXT87 TEXT56 TEXT47 TEXT77 TEXT20 TEXT25 TEXT28 TEXT17 TEXT12 TEXT11

text22 text23 text24 text25 text26 text27 text28 text29 text30 text31 text32 text33 text34 text35 text36 text37 tex

1 TEXT89 TEXT12 TEXT47 TEXT70 TEXT5 TEXT82 TEXT31 TEXT98 TEXT63 TEXT8 TEXT99 TEXT72 TEXT25 TEXT18 T

2 TEXT27 TEXT89 TEXT81 TEXT42 TEXT1 TEXT24 TEXT18 TEXT10 TEXT98 TEXT26 TEXT66 TEXT95 TEXT41 TEXT63 T

text41 text42 text43 text44 text45 text46 text47 text48 text49 text50 text51 text52 text53 text54 text55 text56 te

1 TEXT92 TEXT11 TEXT45 TEXT33 TEXT76 TEXT52 TEXT79 TEXT77 TEXT90 TEXT57 TEXT27 TEXT94 TEXT86 TEXT43

2 TEXT30 TEXT64 TEXT9 TEXT43 TEXT68 TEXT72 TEXT49 TEXT31 TEXT33 TEXT15 TEXT84 TEXT83 TEXT40 TEXT51

text60 text61 text62 text63 text64 text65 text66 text67 text68 text69 text70 text71 text72 text73 text74 text75 te

1 TEXT38 TEXT7 TEXT51 TEXT53 TEXT75 TEXT87 TEXT93 TEXT44 TEXT68 TEXT39 TEXT85 TEXT6 TEXT97 TEXT88 T

2 TEXT35 TEXT79 TEXT80 TEXT100 TEXT48 TEXT46 TEXT65 TEXT55 TEXT52 TEXT61 TEXT85 TEXT38 TEXT6 TEXT93

text79 text80 text81 text82 text83 text84 text85 text86 text87 text88 text89 text90 text91 text92 text93 text94 te

1 TEXT58 TEXT3 TEXT40 TEXT17 TEXT62 TEXT61 TEXT29 TEXT56 TEXT67 TEXT42 TEXT36 TEXT48 TEXT41 TEXT64 T

2 TEXT3 TEXT37 TEXT14 TEXT16 TEXT22 TEXT67 TEXT76 TEXT13 TEXT45 TEXT73 TEXT86 TEXT62 TEXT96 TEXT92 T

text98 text99 text100

1 TEXT28 TEXT22 TEXT100

2 TEXT53 TEXT99 TEXT21

[reached 'max' / getOption("max.print") -- omitted 853 rows]

Demographic Information and Data Cleaning

```
> fc<-fc %>% filter(SO %in% c(1, 3))
> fc<-fc %>% filter(Gender %in% c(1, 2) & Gender == Sex)
> unique(fc$ResponseId)
```

589 participants

```
> table(fc$Gender)
```

1 2

280 312

280 men, 312 women

```
> table(fc$SO)
```

```
  1   3
479 113
```

479 heterosexual, 113 bisexual*(see paper or Variable Coding Key)

```
> mean(as.numeric(fc$Age), na.rm=TRUE)
[1] 25.21489
```

```
> table(fc$dating_exp)

  1   2   3
340 243   9
```

340 experience with dating apps, 243 no experience, 9 (unknown meaning, probably other or prompt)

```
> table(fc$Status.1)

  1   2   3
315 262  15
```

315 single, 262 in a relationship, 15 other(see above)

Reformatting

```
> fc_long<-pivot_longer(fc, cols = starts_with("X"),
+                       names_to = "rating",
+                       names_prefix = "X",
+                       values_to = "rate",
+                       values_drop_na = TRUE,
+                       cols_vary = "fastest")
```

Reformat data to long format. Organize ratings.

```
> head(fc_long)
# A tibble: 6 × 260

  ResponseId   Progress Gender  Sex  Age  SO dating_exp Status.1 BFI1_1 BFI1_2 BFI1_3 BFI1_4
BFI1_5 BFI1_6 BFI1_7 BFI1_8 BFI1_9 BFI1_10

  <chr>        <int>  <int> <int> <int> <int>   <int>   <int>  <int>  <int>  <int>  <int>  <int>
<int>  <int>  <int>  <int>

1 R_BLKJUB3sKl...   100   1   1   21   1    1   2   5   2   5   4   4   3   3   5   2   5
2 R_BLKJUB3sKl...   100   1   1   21   1    1   2   5   2   5   4   4   3   3   5   2   5
3 R_BLKJUB3sKl...   100   1   1   21   1    1   2   5   2   5   4   4   3   3   5   2   5
4 R_BLKJUB3sKl...   100   1   1   21   1    1   2   5   2   5   4   4   3   3   5   2   5
5 R_BLKJUB3sKl...   100   1   1   21   1    1   2   5   2   5   4   4   3   3   5   2   5
6 R_BLKJUB3sKl...   100   1   1   21   1    1   2   5   2   5   4   4   3   3   5   2   5
```

ℹ 242 more variables: BFI1_11 <int>, BFI1_12 <int>, BFI1_13 <int>, BFI1_14 <int>, BFI1_15 <int>, BFI1_16 <int>, BFI1_17 <int>,

BFI1_18 <int>, BFI1_19 <int>, BFI1_20 <int>, BFI1_21 <int>, BFI1_22 <int>, BFI1_23 <int>, BFI1_24 <int>, BFI1_25 <int>, BFI1_26 <int>,

BFI1_27 <int>, BFI1_28 <int>, BFI1_29 <int>, BFI1_30 <int>, BFI1_31 <int>, BFI1_32 <int>, BFI1_33 <int>, BFI1_34 <int>, BFI1_35 <int>,

BFI1_36 <int>, BFI1_37 <int>, BFI1_38 <int>, BFI1_39 <int>, BFI1_40 <int>, BFI1_41 <int>, BFI1_42 <int>, BFI1_43 <int>, BFI1_44 <int>,

Q10 <int>, Desc_or_pic <int>, Attention <int>, Q20 <int>, Q21 <chr>, Random.ID <int>, image1 <chr>, image2 <chr>, image3 <chr>,

image4 <chr>, image5 <chr>, image6 <chr>, image7 <chr>, image8 <chr>, image9 <chr>, image10 <chr>, image11 <chr>, image12 <chr>,

image13 <chr>, image14 <chr>, image15 <chr>, image16 <chr>, image17 <chr>, image18 <chr>, image19 <chr>, image20 <chr>, …

ℹ Use `colnames()` to see all variable names

> ## current image: rating Q[nr] -> image[nr]

```
>fc_long <- fc_long %>%
 + rowwise() %>%
 + mutate(
+    image_number = (str_extract(rating, "^\\d+")),
+    current_image = 
c_across(starts_with("image"))[as.numeric(image_number)]
+  ) %>%
+  ungroup()

>fc_long <- fc_long %>%
+  rowwise() %>%
+  mutate(
+    text_number = (str_extract(rating, "^\\d+")),
+    current_text = c_across(starts_with("text"))[as.numeric(text_number)]
+  ) %>%
+  ungroup()
```

Makes two new variables, current_image and current_text which contain the image and text the participant rates for that trial respectively.

```
fc_long_summary <- fc_long %>%
  group_by(ResponseId, current_image, current_text) %>%
  summarise(
    across(
```

```
      c(rate),
      ~ mean(.x, na.rm = TRUE)
    ),
    .groups = "drop"
  )
```

Makes a summary, containing only the variables relevant for the final analysis.

filter

Model Comparisons

```
LMEmodel<- lmer(rate ~ (1|current_image) + (1|ResponseId) +
(1|current_text), data = fc_long_summary)
```

```
LMEmodelcor<-lmer(rate ~ (1|current_image) + (1|ResponseId) +
(1|current_text) + (1|current_text:current_image), data = fc_long_summary)
```

```
summary(LMEmodel)
```

Linear mixed model fit by REML ['lmerMod']

Formula: rate ~ (1 | current_image) + (1 | ResponseId) + (1 | current_text)

Data: fc_long_summary

REML criterion at convergence: 56186.6

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.8561	-0.8314	-0.1326	0.8764	3.1848

Random effects:

Groups	Name	Variance	Std.Dev.
ResponseId	(Intercept)	0.047197	0.21725
current_image	(Intercept)	0.006831	0.08265
current_text	(Intercept)	0.014030	0.11845
Residual		0.178093	0.42201

Number of obs: 48271, groups: ResponseId, 531; current_image, 199; current_text, 100

Fixed effects:

Estimate Std. Error t value

(Intercept) 1.43609 0.01641 87.5

#control for the interaction effect by comparing the full model to a nested model

```
> anova(LMEmodelcor, LMEmodel)
```

refitting model(s) with ML (instead of REML)

Data: fc_long_summary

Models:

LMEmodel: rate ~ (1 | current_image) + (1 | ResponseId) + (1 | current_text)

LMEmodelcor: rate ~ (1 | current_image) + (1 | ResponseId) + (1 | current_text) + (1 | current_text:current_image)

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)	
LMEmodel	5	56190	56234	-28090	56180				
LMEmodelcor	6	56188	56241	-28088	56176	4.0119	1	0.04518	*

Signif. codes: 0 '*' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1**

```
# other factors
>model_yes_text <- lmer(
  rate ~ +
    (1 | current_image) +
    (1 | current_text) +
    (1 | ResponseId),
  data = fc_long_summary,
  REML = FALSE
)

>model_no_text <- lmer(
  rate ~ +
    (1 | ResponseId) +
    (1 | current_image),
 data = fc_long_summary,
  REML = FALSE
)

anova(model_yes_text, model_no_text)
```

Data: fc_long_summary

Models:

model_no_text: rate ~ +(1 | ResponseId) + (1 | current_image)

model_yes_text: rate ~ +(1 | current_image) + (1 | current_text) + (1 | ResponseId)

```
               npar   AIC   BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
model_no_text     4 59427 59463 -29710     59419
model_yes_text    5 56190 56234 -28090     56180 3239.2  1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
#responseid

>model_yes_id <- lmer(
  rate ~ +
    (1 | current_image) +
    (1 | current_text) +
    (1 | ResponseId),
  data = fc_long_summary,
  REML = FALSE
)

>model_no_id <- lmer(
  rate ~ +
    (1 | current_text) +
    (1 | current_image),
  data = fc_long_summary,
  REML = FALSE
)

>anova(model_yes_id, model_no_id) #significant
```

Data: fc_long_summary

Models:

model_no_id: rate ~ +(1 | current_text) + (1 | current_image)

model_yes_id: rate ~ +(1 | current_image) + (1 | current_text) + (1 | ResponseId)

```
             npar   AIC   BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
```

model_no_id 4 65387 65422 -32690 65379

model_yes_id 5 56190 56234 -28090 56180 9198.8 1 < 2.2e-16 ***

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#significance of image random effect, model comparison

```
>model_yes_image <- lmer(
  rate ~ +
    (1 | current_image) +
    (1 | current_text) +
    (1 | ResponseId),
  data = fc_long_summary,
  REML = FALSE
)

>model_no_image <- lmer(
  rate ~ +
    (1 | ResponseId) +
    (1 | current_text),
  data = fc_long_summary,
  REML = FALSE
)
```

#calculate the variance components

```
>VarCorr(LMEmodel) #variance components
```

#proportions

```
>vc <- as.data.frame(VarCorr(LMEmodel))

>vc$prop <- vc$vcov / sum(vc$vcov)

>vc
```

grp var1 var2 vcov sdcor prop

1 ResponseId (Intercept) <NA> 0.047197151 0.21724905 0.19174115

2 current_image (Intercept) <NA> 0.006830649 0.08264774 0.02774991

3 current_text (Intercept) <NA> 0.014029951 0.11844809 0.05699749

4 Residual <NA> <NA> 0.178092593 0.42201018 0.72351145

#prop: 19.17% Subject Random Effect

 # 2.78% Image

 # 5.70% Text

 # 72.35% Residual

```
>anova(model_yes_image, model_no_image) #significant
```

```
Data: fc_long_summary
Models:
model_no_image: rate ~ +(1 | ResponseId) + (1 | current_text)
model_yes_image: rate ~ +(1 | current_image) + (1 | current_text) + (1 | ResponseId)
                npar   AIC   BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
model_no_image     4 57481 57516 -28736     57473
model_yes_image    5 56190 56234 -28090     56180 1292.5  1  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Additional Analyses

```
> icc(LMEmodel)
```

```
# Intraclass Correlation Coefficient

    Adjusted ICC: 0.276
  Unadjusted ICC: 0.276
```